

Coordination Compounds

The two school note sets, combined and enlarged

Part	Source file	Source pages	Sheets here
Solved multiple-choice questions	Coordination chemistry.pdf	20	20
Case-study questions	Coordination compound.pdf	34	34

What was done. Both note sets are image-only and their page images are small, so the type is cramped. **Nothing here has been retyped or redrawn** — these are the original pages. What changed is how much of the sheet each one gets: the blank margins are cropped away, a page set in two columns is split at its gutter so each column gets a sheet of its own, and what remains is enlarged, sharpened and contrast-normalised onto A4.

Nothing is lost. Only blank paper is cropped, and every output pixel comes from a source pixel. The build measures the ink in each source page against the ink kept, and refuses to write the document if any is missing.

Part 1 · Solved multiple-choice questions

Twenty worked multiple-choice questions, each with its solution.

1. The sum of primary valance and secondary valance of the metal M in the complex $[M(en)_2(Ox)]Cl$ is L

a) 3

b) 6

c) -3

d) 9

Solution

In the complex $[M(en)_2(Ox)]Cl$ For the central metal ion M^{3+}

The primary valance is $= +3$

The secondary valance $= 6$

sum of primary valance and secondary valance $= 3+6 = 9$

Answer : option (d)

2. An excess of silver nitrate is added to 100ml of a 0.01M solution of pentaqua chloride chromium(III)chloride. The number of moles of AgCl precipitated would be

a) 0.02

b) 0.002

c) 0.01

d) 0.2

Solution

The complex is $[M(H_2O)_5 Cl] Cl_2$

1000 ml of 1M solution of the complex gives 2 moles of Cl^- ions

1000 ml of 0.01M solution of the complex will give

$[100 \text{ ml} \times 0.01M \times 2Cl^-] / [1000 \text{ ml} \times 1M] = 0.002 \text{ moles of } Cl^- \text{ ions}$

Answer : option (b)

3. A complex has a molecular formula $\text{MSO}_4 \cdot \text{Cl} \cdot 6\text{H}_2\text{O}$. The aqueous solution of it gives white precipitate with Barium chloride solution and no precipitate is obtained when it is treated with silver nitrate solution. If the secondary valence of the metal is six, which one of the following correctly represents the complex?

- a) $[\text{M}(\text{H}_2\text{O})_4\text{Cl}]\text{SO}_4 \cdot 2\text{H}_2\text{O}$
- b) $[\text{M}(\text{H}_2\text{O})_6]\text{SO}_4$
- c) $[\text{M}(\text{H}_2\text{O})_5\text{Cl}]\text{SO}_4 \cdot \text{H}_2\text{O}$**
- d) $[\text{M}(\text{H}_2\text{O})_3\text{Cl}]\text{SO}_4 \cdot 3\text{H}_2\text{O}$

Solution

Molecular formula: $\text{MSO}_4\text{Cl} \cdot 6\text{H}_2\text{O}$.

Formation of white precipitate with Barium chloride indicates that SO_4^{2-} ions are outside the coordination sphere, and no precipitate with AgNO_3 solution indicates that the Cl^- ions are inside the coordination sphere. Since the coordination number of M is 6, Cl^- and 5 H_2O are ligands, remaining 1 H_2O molecular and SO_4^{2-} are in the outer coordination sphere.

Answer : option (c)

4. Oxidation state of Iron and the charge on the ligand NO in $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ are

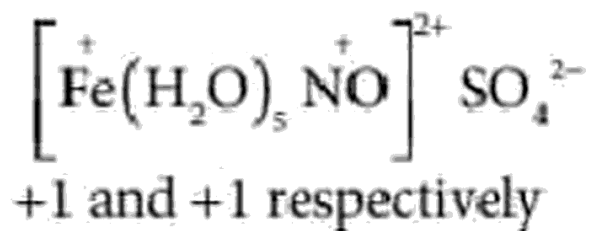
a) +2 and 0 respectively

b) +3 and 0 respectively

c) +3 and -1 respectively

d) +1 and +1 respectively

Solution



Answer : option(d)

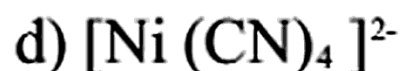
5. As per IUPAC guidelines, the name of the complex $[\text{Co}(\text{en})_2(\text{ONO})\text{Cl}]\text{Cl}$ is

- a) chlorobisethylenediaminenitritocobalt(III) chloride
- b) chloridobis(ethane-1,2-diamine)nitro *k*-Ocobaltate(III) chloride
- c) chloridobis(ethane-1,2-diammine)nitrito *k*-Ocobalt(II) chloride
- d) chloridobis(ethane-1,2-diamine)nitro *k*-Ocobalt(III) chloride**

6. IUPAC name of the complex $\text{K}_3[\text{Al}(\text{C}_2\text{O}_4)_3]$ is

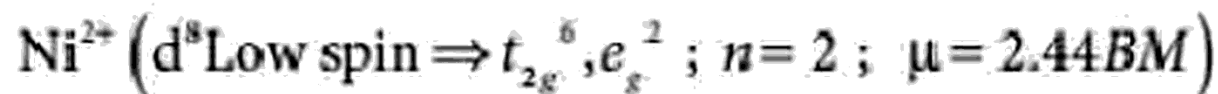
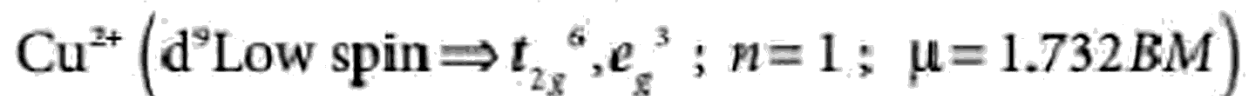
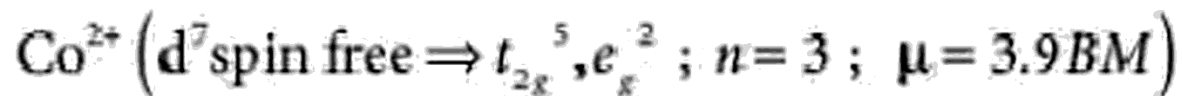
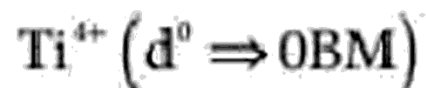
- a) potassiumtrioxalatoaluminium(III)
- b) potassiumtrioxalatoaluminate(II)
- c) potassiumtrisoxalatoaluminate(III)
- d) potassiumtrioxalatoaluminate(III)**

7. A magnetic moment of 1.73BM will be shown by one among the following



Solution

Answer : option(c)



8. Crystal field stabilization energy for high spin d^5 octahedral complex is

a) $-0.6 \Delta_0$.

b) 0

c) $2(P - \Delta_0)$

d) $2(P + \Delta_0)$

Solution

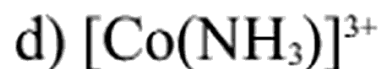
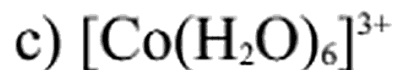
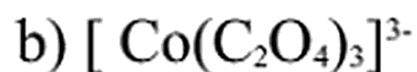
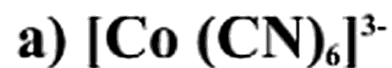
The electronic configuration t_{2g}^3, e_g^2

$$[3 \times (-0.4) + 2(0.6)]\Delta_0$$

$$[-1.2 + 1.2]\Delta_0 = 0$$

Answer : option(b)

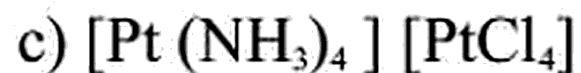
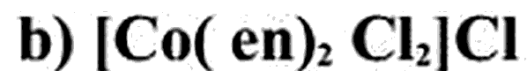
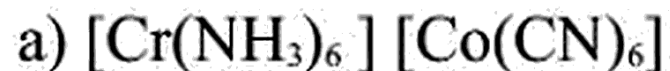
9. In which of the following coordination entities the magnitude of Δ_o will be maximum?



Solution

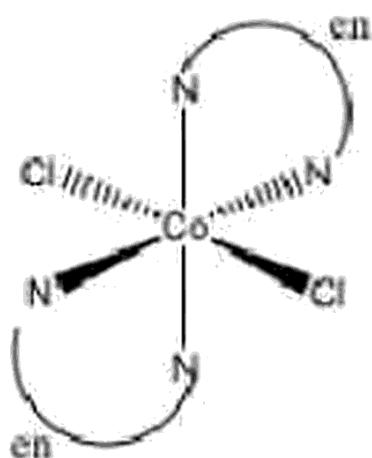
In all the complexes, the central metal ion is Co^{3+} , among the given ligands CN^- is the strongest ligand, which causes large crystal field splitting i.e maximum Δ_o

10. Which one of the following will give a pair of enantiomorphs?

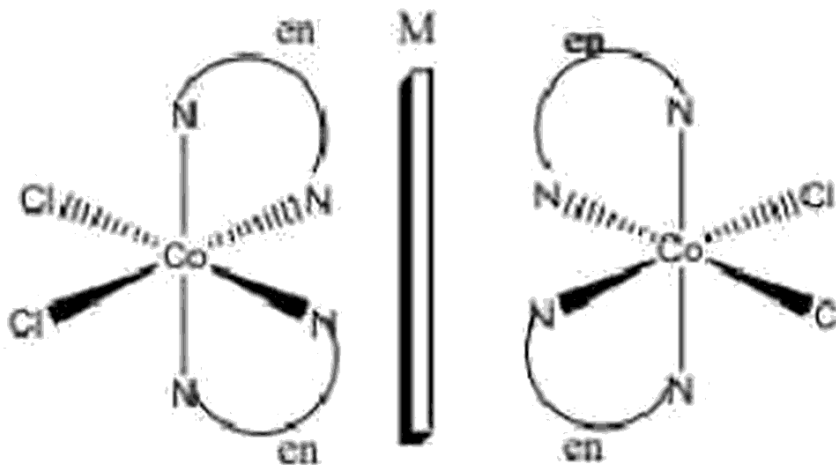


Solution

Answer : option(b)



inactive



Active forms (enantiomorphs)

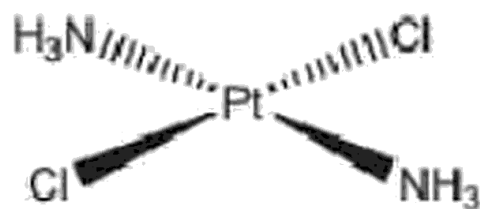
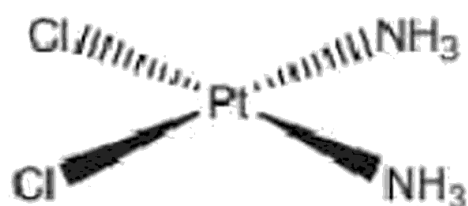
Complexes given in other options (a), (c) and (d) have symmetry elements and hence they are optically inactive.

11. Which type of isomerism is exhibited by $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$?

- a) Coordination isomerism
- b) Linkage isomerism
- c) Optical isomerism
- d) Geometrical isomerism**

Solution

Answer : option(d)



12. How many geometrical isomers are possible for $[\text{Pt}(\text{Py})(\text{NH}_3)(\text{Br})(\text{Cl})]$?

a) 3

b) 4

c) 0

d) 15

Solution

Three isomers. If we consider any one of the ligands as reference (say Py), the arrangement of other three ligands (NH_3 , Br and Cl^-) with respect to (Py) gives three geometrical isomers.

13. Which one of the following pairs represents linkage isomers?

a) $[\text{Cu}(\text{NH}_3)_4][\text{PtCl}_4]$ and $[\text{Pt}(\text{NH}_3)_4][\text{CuCl}_4]$

b) $[\text{Co}(\text{NH}_3)_5(\text{NO}_3)]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})]\text{SO}_4$

c) $[\text{Co}(\text{NH}_3)_4(\text{NCS})_2]\text{Cl}$ and $[\text{Co}(\text{NH}_3)_4(\text{SCN})_2]\text{Cl}$

d) both (b) and (c)

Solution

(a) coordination isomers

(b) no isomerism (different molecular formula)

(c) $\leftarrow$ NCS , $\leftarrow$ SCN coordinating atom differs : linkage isomers

14. Which kind of isomerism is possible for a complex $[\text{Co} (\text{NH}_3)_4 \text{Br}_2] \text{Cl}$?

a) geometrical and ionization

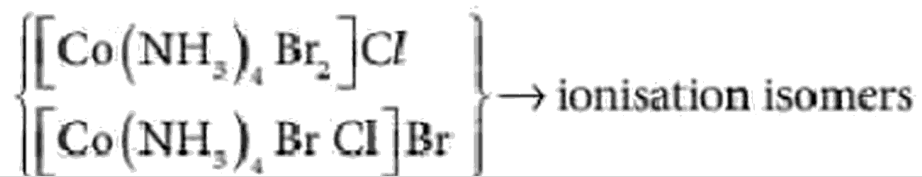
b) geometrical and optical

c) optical and ionization

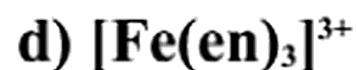
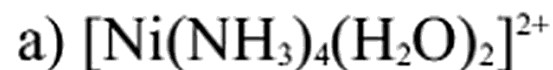
d) geometrical only

Solution

For $[\text{MA}_4 \text{B}_2]^{n+}$ complexes-geometrical isomerism is possible



15. Which one of the following complexes is not expected to exhibit isomerism?



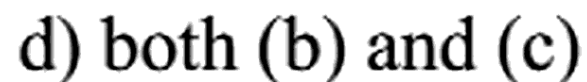
Solution

Option (a) and (b) – geometrical isomerism is possible

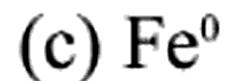
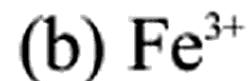
Option (c) – ionization isomerism is possible

Option (d) – no possibility to show either constitutional isomerism or stereo isomerism

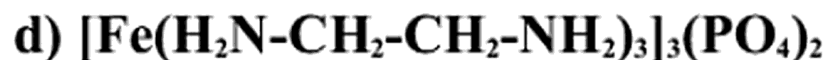
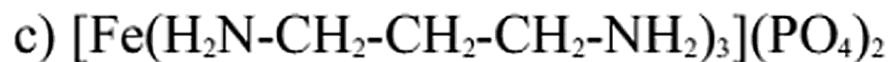
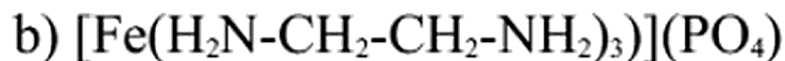
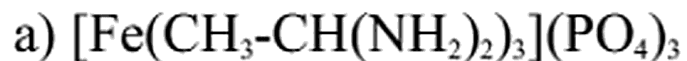
16. A complex in which the oxidation number of the metal is zero is



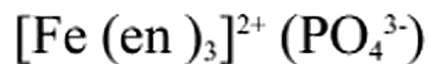
Solution



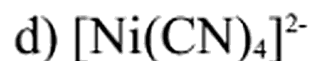
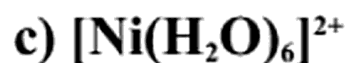
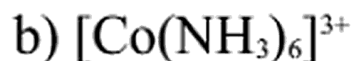
17. Formula of tris(ethane-1,2-diamine)iron(II)phosphate



Solution



18. Which of the following is paramagnetic in nature?



Solution

(a) Zn^{2+} ($d^{10} \Rightarrow$ diamagnetic)

(b) Co^{3+} (d^6 Low spin $\Rightarrow t_2g^6, e_g^0$ diamagnetic)

(c) Ni^{2+} (d^8 Low spin $\Rightarrow t_2g^6, e_g^2$ paramagnetic)

(d) $[\text{Ni}(\text{CN})_4]^{2-}$ (dsp^2 ; square planar, diamagnetic)

19. Fac-mer isomerism is shown by

a) $[\text{Co}(\text{en})_3]^{3+}$

b) $[\text{Co}(\text{NH}_3)_4(\text{Cl})_2]^+$

c) $[\text{Co}(\text{NH}_3)_3(\text{Cl})_2]$

d) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$

Solution

$[\text{Co}(\text{NH}_3)_3(\text{Cl})_3]$

20. Choose the correct statement.

- a) Square planar complexes are more stable than octahedral complexes
- b) The spin only magnetic moment of $[\text{Cu}(\text{Cl})_4]^{2-}$ is 1.732 BM and it has square planar structure.
- c) Crystal field splitting energy (Δ_0) of $[\text{FeF}_6]^{4-}$ is higher than the (Δ_0) of $[\text{Fe}(\text{CN})_6]^{4-}$
- d) crystal field stabilization energy of $[\text{V}(\text{H}_2\text{O})_6]^{2+}$ is higher than the crystal field stabilization of $[\text{Ti}(\text{H}_2\text{O})_6]^{2+}$**

Solution

$$V^{2+} (t_2g^3, e_g^0; CFSE = 3 \times (-0.4)\Delta_0 = -1.2 \Delta_0)$$

$$Ti^{2+} (t_2g^2, e_g^0; CFSE = 2 \times (-0.4)\Delta_0 = -0.8 \Delta_0)$$

Statements given in option (a), (b), and (c) are wrong.

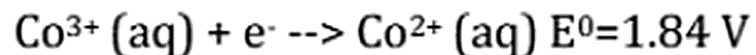
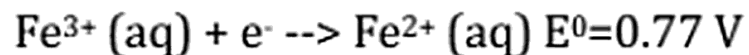
The correct statements are

(a) since, the crystal field stabilization is more in octahedral field, octahedral complexes are more stable than square planar complexes.

$$[FeF_6]^{4-} \left(\begin{array}{l} Fe^{2+} - d^6 \text{ High Spin } - t_{2g}^4, e_g^2 \\ CFSE = 4 \times (-0.4) + (0.6) \times 2 + P \end{array} \right)$$

$$[Fe(CN)_6]^{4-} \left(\begin{array}{l} Fe^{2+} - d^6 \text{ Low Spin } - t_{2g}^6, e_g^0 \\ CFSE = 6 \times (-0.4) + 3P \end{array} \right)$$

6. The following reduction potentials are known for various 3+ first-row transition metal ions in aqueous solution:



Explain why the reduction potential for Fe^{3+} is abnormally low.

H_2O is a weak-field ligand, thus, in all the hexaaqua octahedral species of the above ions, the metal is in a high-spin state. Mn^{3+} is d^4 , Fe^{3+} is d^5 , Co^{3+} is d^6 . Only Fe^{3+} has all its d orbitals half-filled which are a relatively stable electronic configuration. Thus, the reduction potential of the Fe (III) ion is less than its neighbors in the periodic table.

7. For each of the following ions, show the d orbital occupancies in both weak and strong octahedral fields. Indicate the total number of unpaired electrons in each case.

(a) Mn^{2+} (b) Zn^{2+} (c) Cr^{3+} (d) Mn^{3+} (e) Co^{3+}

(a) $\text{Mn}^{2+} d^5$

Weak field ($t_{2g}^3e_g^2$) 5 unpaired electrons

Strong field ($t_{2g}^5e_g^0$) 1 unpaired electron

(b) Zn^{2+} d^{10}

Weak and strong field ($t_{2g}^6e_g^4$) 0 unpaired electrons

(c) Cr^{3+} d^3

Weak and strong field ($t_{2g}^3e_g$) 3 unpaired electrons

(d) Mn^{3+} d^4

Weak field ($t_{2g}^3e_g^1$) 4 unpaired electrons

Strong field ($t_{2g}^4e_g^0$) 2 unpaired electrons

(e) Co^{3+} d^6

Weak field ($t_{2g}^4e_g^2$) 4 unpaired electrons

Strong field ($t_{2g}^6e_g^0$) 0 unpaired electrons

Part 2 · Case-study questions

Thirteen case studies with their questions, then the answers, then the hints and explanations.

Coordination Compounds

Case Study Based Questions

Case Study 1

Co-ordination compounds show structural isomerism and it is different from the nature of the isomerism which we generally come across in organic compounds. The structural isomers may be classified as ionisation, hydrate, co-ordination and linkage isomers depending upon their nature. The electrical conductivity of these isomers depends upon the number of ions which they furnish upon dissociation in aqueous solution. The isomers belonging to a particular type can also be distinguished from each other with the help of certain tests. In addition to the structural isomerism, the co-ordination compounds are also involved in geometrical and optical isomerism. Whereas optical isomerism is normally shown by octahedral complexes, both octahedral and square planar complexes can exhibit geometrical isomerism.

Read the given passage carefully and give the answer of the following questions:

Q1. An aqueous solution of CoCl_2 on addition of excess of concentrated HCl turns blue due to the formation of:

- a. $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]$
- b. $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_4]^{2-}$
- c. $(\text{CoCl}_4)^{2-}$
- d. $[\text{Co}(\text{H}_2\text{O})_2\text{Cl}_2]$

Q2. Which one is the most likely structure of $\text{CrCl}_6 \cdot 6\text{H}_2\text{O}$ if $1/3$ of the total chlorine in the compounds is precipitated by adding AgNO_3 solution?

- a. $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$
- b. $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot (\text{H}_2\text{O})_3$
- c. $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2] \cdot 2\text{H}_2\text{O}$
- d. $(\text{Cr}(\text{H}_2\text{O})_5\text{Cl})\text{Cl}_2 \cdot 2\text{H}_2\text{O}$

Q3. One mole of the complex compound $\text{Co}(\text{NH})_5\text{Cl}_3$ gives 3 mole of ions on dissolution in water. One mole of the same complex reacts with two moles of AgNO_3 to yield two moles of AgCl(s) . The structure of the complex is:

- a. $[\text{Co}(\text{NH}_3)_3\text{Cl}_3] \cdot 2\text{NH}_3$ b. $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{ClNH}_3$
 c. $[\text{Co}(\text{NH}_3)_4\text{Cl}]\text{Cl}_2 \cdot \text{NH}_3$ d. $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$

Q4. Which of the following will not show geometrical isomerism?

- a. $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ b. $[\text{CoCl}_2(\text{en})_2]\text{Cl}$
 c. $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ d. $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$.

Answers

1. (c) $(\text{CoCl}_4)^{2-}$
2. (c) $(\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2)\text{Cl} \cdot 2\text{H}_2\text{O}$
3. (d) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
4. (c) $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$

Case Study 2

Metal complexes show different colours due to d-d transitions. The complex absorbs light of specific wavelength to promote the electron from t_2 to e level. The colour of the complex is due to the transmitted light, which is complementary of the colour absorbed.

The wave number of light absorbed by different complexes of Cr ion are given below:

Complex	Wave number of light absorbed (cm^{-1})	Energy of light absorbed (kJ/mol)
$[\text{CrA}_6]^{3-}$	13640	163
$[\text{CrB}_6]^{3+}$	17830	213
$[\text{CrC}_6]^{3+}$	21680	259
$[\text{CrD}_6]^{3-}$	26280	314

Read the given passage carefully and give the answer of the following questions: (CBSE SQP 2023-24)

Q1. Out of the ligands 'A', 'B', 'C' and 'D', which ligand causes maximum crystal field splitting? Why?

OR

Which of the two, 'A', or 'D' will be a weak field ligand? Why?

Q2. Which of the complexes will be violet in colour? [CrA] or [CrB]³⁺ and why? (Given: If 560-570 nm of light is absorbed, the colour of the complex observed is violet.)

Q3. If the ligands attached to Cr³⁺ ion in the complexes given in the table above are water, cyanide ion, chloride ion and ammonia (not in this order) Identify the ligand, write the formula and IUPAC name of the following:

(i) [CrA]³⁺

(ii) [CrC₂]³⁺

Answers

1. 'D' causes maximum crystal field splitting. Energy is directly proportional to the wave number. Maximum energy of light is required for an electron to jump from t_{2g} to e_g in case of (CrD₆)³⁻

OR

'A' will be a weak field ligand. The splitting caused is least in this case as the energy required for electron to jump from t_{2g} to e_g is minimum.

2. $[\text{CrB}_3]^+$ will be violet in colour because wavelength of light absorbed is $1/17830-560\text{nm}$ for the complex while $1/13640733\text{nm}$ for $(\text{CrA}_6)_3^-$ complex.

3. (i) $(\text{CrCl}_6)^{3+}$ Hexachloridochromate (III) ion

(ii) $(\text{Cr}(\text{NH}_3)_6)^{3+}$, Hexaamminechromium (III) ion

A=Cl, B=H₂O, C = NH₃, D = CN

Case Study 3

In coordination compounds, metals show two types of linkages, primary and secondary. Primary valencies are ionisable and are satisfied by negatively charged ions. Secondary valencies are non-ionisable and are satisfied by neutral or negative ions having lone pair of electrons. Primary valencies are non-directional while secondary valencies decide the shape of the complexes.

Read the given passage carefully and give the answer of the following questions: (CBSE 2023)

Q1. If $\text{PtCl}_2 \cdot 2\text{NH}_3$ does not react with AgNO_3 , what will be its formula?

Q2. What is the secondary valency of $[\text{Co}(\text{en})_3]^+$?

Q3. (i) Write the formula of Iron (III)

Hexacyanidoferrate (II).

(ii) Write the IUPAC name of $[\text{Co}(\text{NH}_3)_5\text{Cl}] \text{Cl}_2$.

OR

**Write the hybridisation and magnetic
behaviour of $[\text{Ni}(\text{CN})_4]^{2-}$**

[Atomic number: Ni = 28]

Answers

1. If $\text{PtCl}_2 \cdot 2\text{NH}_3$ does not react with AgNO_3 , then the formula of the compound would be $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$.

2. The secondary valency of $[\text{Co}(\text{en})_3]^{3+}$ is 6 as there are 4 NH_3 molecules and 2 C^- ions linked to the cobalt atom.

3. (1) The formula is $\text{Fe}_4 [\text{Fe}(\text{CN})_6]_3$

(ii) The IUPAC name of $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ is pentaammine chlorido cobalt (III) chloride.

OR

In the presence of strong field CN ions, all the electrons are paired up. The empty 3d, 3s and two 4p orbitals undergo dsp^2 hybridisation to make bonds with CN-ligands in square planar geometry. Hence, $[\text{Ni}(\text{CN})_4]^{2-}$ is diamagnetic. Coordination compounds are widely present in the minerals, plant and animal worlds and are known to play many important functions in the area of analytical chemistry, metallurgy, biological systems and medicine. Alfred Werner's theory postulated the use of two types of linkages (primary and secondary), by a metal atom/ion in a coordination compound. He predicted the geometrical shapes of a large number of coordination entities using the property of isomerism. The Valence Bond Theory (VBT) explains the formation, magnetic behaviour and geometrical shapes of coordination compounds. It, however, fails to describe the optical properties of these compounds. The Crystal Field Theory (CFT) explains the effect of different crystal fields (provided by the ligands taken as point charges) on the degeneracy of d-orbital energies of the central metal atom/ion.

Read the given passage carefully and give the answer of the following questions: (CBSE 2023)

Q1. When a coordination compound $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ is mixed with AgNO_3 solution, 2 moles of AgCl are precipitated per mole of the compound. Write the structural formula of the complex and secondary valency for Nickel ion.

Q 2. Write the IUPAC name of the ionisation isomer of $[\text{Co}(\text{NH}_3)_4(\text{SO}_4)] \text{Cl}$.

Q3. Using valence bond theory, predict the geometry and magnetic nature of:

(i) $[\text{Ni}(\text{CO})_4]$

(ii) $[\text{Fe}(\text{CN})_6]^{3-}$

[Atomic number: $\text{Ni} = 28$, $\text{Fe} = 26$]

OR

Give reasons:

(i) Low spin tetrahedral complexes are not formed.

(ii) $[\text{Co}(\text{NH}_3)_6]^{3+}$ is an inner orbital complex where as

[Ni(NH₃)₂]²⁺ is an outer orbital complex.

[Atomic number: Co = 27, Ni = 28]

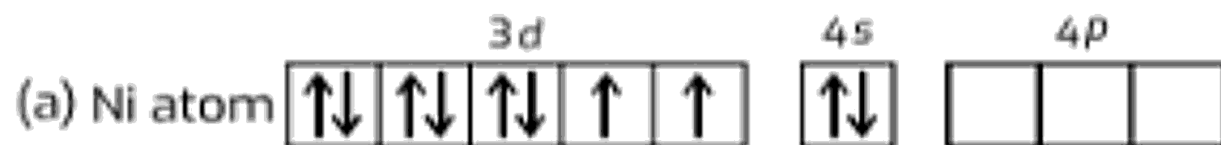
Answers

1. The structural formula of the complex is [Ni(H₂O)₆]Cl₂ and the secondary valency for nickel ion is 6 in it.

2. The IUPAC name of the ionisation isomer of [Co(NH₃)₅(SO₄)]Cl is pentaammine sulphato cobalt (III) chloride.

3. (i) In [Ni(CO)₄], oxidation state of Ni is zero.

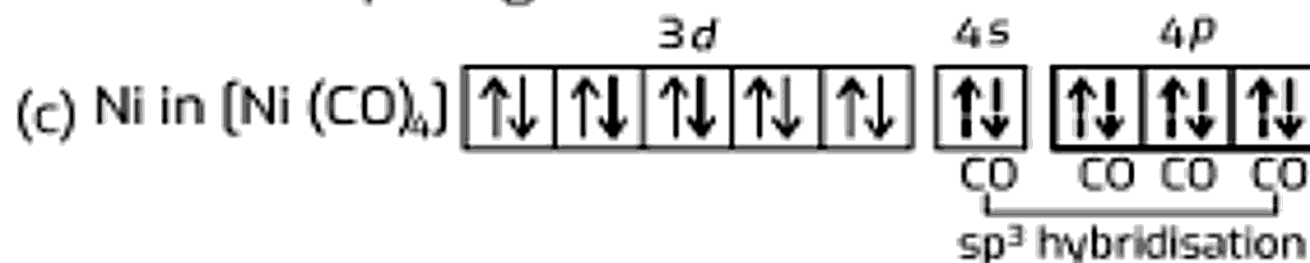
²⁸Ni = [Ar] 3d⁸ 4s²



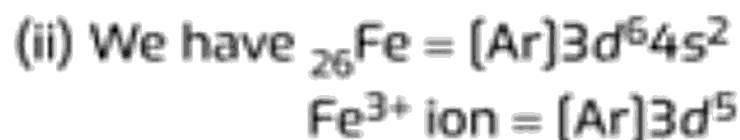
(b) Ni atom after the rearrangement in presence of strong CO ligands.



Co causes pairing of 4 electrons into 3d-orbitals.



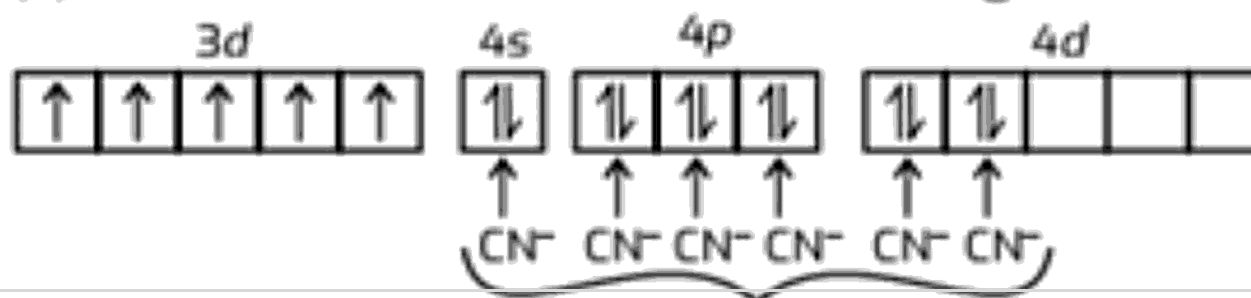
It is diamagnetic as all the electrons present in it are paired.



(a) Fe^{3+} ion



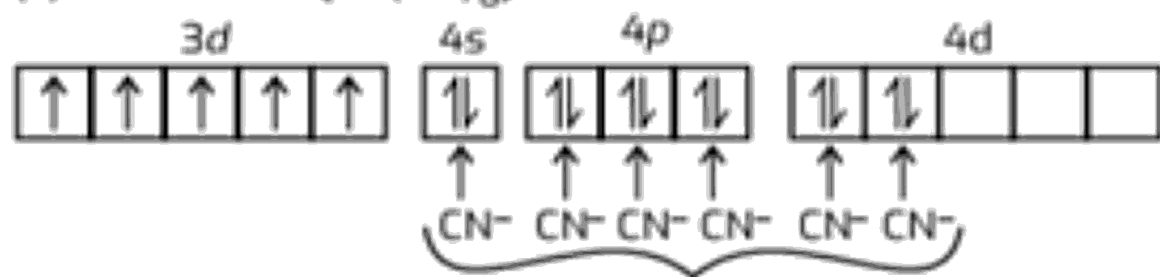
(b) Fe^{3+} ion under influence of weak F^- ligand



sp^3d^2 hybridisation

It is strongly paramagnetic in nature.

(c) Fe^{3+} ion in $[Fe(CN)_6]^{3-}$



Thus, $[Fe(CN)_6]^{3-}$ is strongly paramagnetic in nature with sp^3d^2 hybridisation.

OR

(i) For tetrahedral complexes, the crystal field stabilisation energy is lower than pairing energy. so they are rarely formed in low spin state.

(ii) As inner d-electrons are involved in the complex $[Co(NH_3)_6]^{3+}$, it is an inner orbital complex.

But outer (4d) electrons are involved in the hybridisation of the complex $[Ni(NH_3)_6]^{3+}$ so, it is an outer orbital complex.

Solutions for Questions 4 to 13 are Given Below

Case Study 4

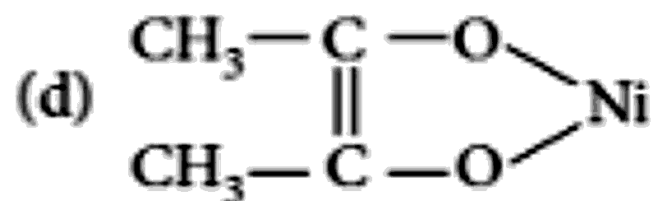
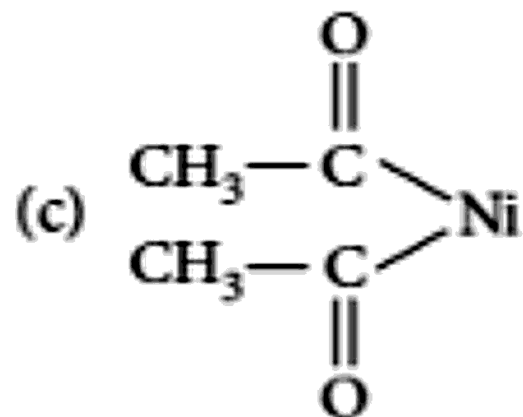
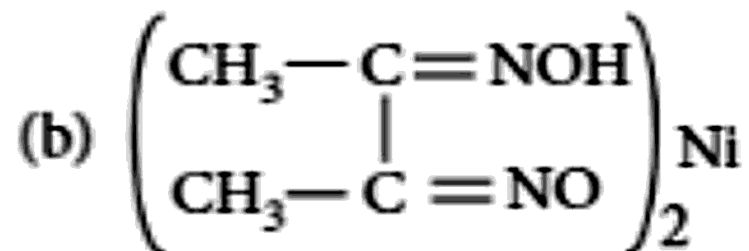
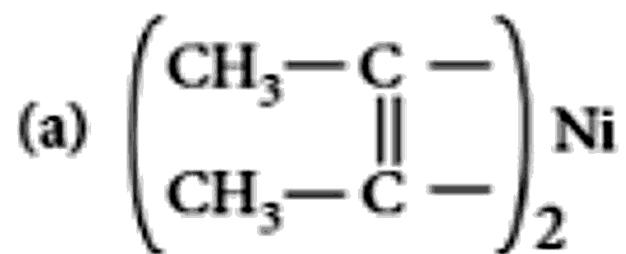
Read the passage given below and answer the following questions :

The molecular compounds which are formed from the combination of two or more simple stable compounds and retain their identity in the solid as well as in the dissolved state are called coordination compounds. Their properties are completely different from the constituents. In coordination compounds, the central metal atom or ion is linked to a number of ions or neutral molecules, called ligands, by coordinate bonds. For example, Dimethyl glyoxime (dmg) is a bidentate ligand chelating large amounts of metals.

When dimethyl glyoxime is added to alcoholic solution of NiCl_2 and ammonium hydroxide is slowly added to it, a rosy red precipitate of a complex is formed.

The following questions are multiple choice questions. Choose the most appropriate answer :

(i) The structure of the complex is



(ii) Oxidation number of Ni in the given complex is

(a) +3

(b) +1

(c) +2

(d) zero.

(iii) Hybridisation and structure of the complex is

(a) sp^3 , tetrahedral

(b) dsp^2 , square planar

(c) sp^3 , square planar

(d) sp^3d , trigonal bipyramidal.

OR

Which of the following is true about this complex?

- (a) It is paramagnetic, containing 2 unpaired electrons.
- (b) It is paramagnetic, containing 1 unpaired electron.
- (c) It is paramagnetic, containing 4 unpaired electrons.
- (d) It is diamagnetic with no unpaired electron.

(iv) Which one will give test for Fe^{3+} ions in the solution?

- | | |
|--|-------------------------------------|
| (a) $[\text{Fe}(\text{CN})_6]^{3-}$ | (b) $[\text{Fe}(\text{CN})_6]^{2-}$ |
| (c) $(\text{NH}_4)_2\text{SO}_4 \cdot \text{FeSO}_4 \cdot 6\text{H}_2\text{O}$ | (d) $\text{Fe}_2(\text{SO}_4)_3$ |

Case Study 5

Read the passage given below and answer the following questions :

Coordination compounds are formulated and named according to the IUPAC system.

Few rules for naming coordination compounds are :

- (I) In ionic complex, the cation is named first and then the anion.
- (II) In the coordination entity, the ligands are named first and then the central metal ion.
- (III) When more than one type of ligands are present, they are named in alphabetical order of preference without any consideration of charge.

The following questions are multiple choice questions. Choose the most appropriate answer :

- (i) The IUPAC name of the complex $[\text{Pt}(\text{NH}_3)_3\text{Br}(\text{NO}_2)\text{Cl}]\text{Cl}$ is
- (a) triamminechlorobromonitroplatinum(IV) chloride
 - (b) triamminebromonitrochloroplatinum(IV) chloride
 - (c) triamminebromidochloridonitroplatinum(IV) chloride
 - (d) triamminenitrochlorobromoplatinum(IV) chloride.
- (ii) The IUPAC name of $[\text{Ni}(\text{CO})_4]$ is
- (a) tetracarbonylnickel(II)
 - (b) tetracarbonylnickel(0)
 - (c) tetracarbonylnickelate(II)
 - (d) tetracarbonylnickelate(0).
- (iii) As per IUPAC nomenclature, the name of the complex $[\text{Co}(\text{H}_2\text{O})_4(\text{NH}_3)_2]\text{Cl}_3$ is
- (a) tetraaquadiamminecobalt(II) chloride
 - (b) tetraaquadiamminecobalt(III) chloride
 - (c) diamminetetraaquacobalt(II) chloride
 - (d) diamminetetraaquacobalt(III) chloride.
- (iv) Which of the following represents correct formula of dichloridobis(ethane-1, 2-diamine)cobalt(III) ion?
- (a) $[\text{CoCl}_2(\text{en})]^{2+}$
 - (b) $[\text{CoCl}_2(\text{en})_2]^{2+}$
 - (c) $[\text{CoCl}_2(\text{en})]^+$
 - (d) $[\text{CoCl}_2(\text{en})_2]^+$

OR

Correct formula of pentaamminenitro-O-cobalt(III) sulphate is

- (a) $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{SO}_4$
- (b) $[\text{Co}(\text{ONO})(\text{NH}_3)_5]\text{SO}_4$
- (c) $[\text{Co}(\text{NO}_2)(\text{NH}_3)_4](\text{SO}_4)_2$
- (d) $[\text{Co}(\text{ONO})(\text{NH}_3)_4](\text{SO}_4)_2$

Case Study 6

Read the passage given below and answer the following questions :

Iron forms many complexes in its +2 and +3 oxidation states such as $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (A); $[\text{Fe}(\text{CN})_6]^{4-}$ (B); $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (C); $[\text{Fe}(\text{CN})_6]^{3-}$ (D), etc., They exhibit different magnetic properties and undergo different hybridisation of iron.

The following questions are multiple choice questions. Choose the most appropriate answer :

(i) Which of the following statements is correct?

- (a) (B) is paramagnetic while (C) is diamagnetic.
- (b) Both (B) and (D) are outer orbital complexes.
- (c) Both (A) and (C) are paramagnetic.
- (d) (A) is outer orbital complex and (C) is inner orbital complex.

(ii) The complex having maximum magnetic moment is

- (a) (A)
- (b) (B)
- (c) (C)
- (d) (D)

OR

Which of the following does not represent correct configuration of the *d*-orbitals in the given complexes?

- (a) (A) : $t_{2g}^4 e_g^2$
- (b) (B) : $t_{2g}^6 e_g^0$
- (c) (C) : $t_{2g}^4 e_g^1$
- (d) (D) : $t_{2g}^5 e_g^0$

(iii) The spin only magnetic moment of complexes (A), (B), (C) and (D) are respectively (in BM)

- (a) $2\sqrt{6}, 0, \sqrt{35}, \sqrt{3}$
- (b) $0, 2\sqrt{6}, \sqrt{35}, \sqrt{3}$
- (c) $\sqrt{15}, 2\sqrt{6}, \sqrt{3}, 0$
- (d) $\sqrt{3}, \sqrt{8}, 0, \sqrt{15}$

(iv) Which of the given complexes are outer orbital complexes?

- (a) (A) and (B) only (b) (B) and (C) only (c) (A) and (C) only (d) (B) and (D) only

Case Study 7

Read the passage given below and answer the following questions :

To explain bonding in coordination compounds various theories were proposed. One of the important theory was valence bond theory. According to that, the central metal ion in the complex makes available a number of empty orbitals for the formation of coordination bonds with suitable ligands. The appropriate atomic orbitals of the metal hybridise to give a set of equivalent orbitals of definite geometry.

The *d*-orbitals involved in the hybridisation may be either inner *d*-orbitals *i.e.*, $(n-1)d$ or outer *d*-orbitals *i.e.*, nd . For example, Co^{3+} forms both inner orbital and outer orbital complexes, with ammonia it forms $[\text{Co}(\text{NH}_3)_6]^{3+}$ and with fluorine it forms $[\text{CoF}_6]^{3-}$ complex ion.

The following questions are multiple choice questions. Choose the most appropriate answer :

(i) Which of the following is not true for $[\text{CoF}_6]^{3-}$?

- (a) It is paramagnetic. (b) It has coordination number of 6.
(c) It is outer orbital complex. (d) It involves d^2sp^3 hybridisation.

(ii) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (at. no. of Cr = 24) has a magnetic moment of 3.83 B.M. The correct distribution of 3*d*-electrons in the central metal of the complex is

- (a) $3d_{xy}^1, 3d_{x^2-y^2}^1, 3d_{yz}^1$ (b) $3d_{xy}^1, 3d_{yz}^1, 3d_{zx}^1$
(c) $3d_{xy}^1, 3d_{zy}^1, 3d_{z^2}^1$ (d) $3d_{x^2-y^2}^1, 3d_{z^2}^1, 3d_{xz}^1$

OR

Which of the following is true for $[\text{Co}(\text{NH}_3)_6]^{3+}$?

- (a) It is an octahedral, diamagnetic and outer orbital complex.
- (b) It is an octahedral, paramagnetic and outer orbital complex.
- (c) It is an octahedral, paramagnetic and inner orbital complex.
- (d) It is an octahedral, diamagnetic and inner orbital complex.

(iii) The paramagnetism of $[\text{CoF}_6]^{3-}$ is due to

- (a) 3 electrons
- (b) 4 electrons
- (c) 2 electrons
- (d) 1 electron.

(iv) Which of the following is an inner orbital or low spin complex?

- (a) $[\text{Ni}(\text{H}_2\text{O})_6]^{3+}$
- (b) $[\text{FeF}_6]^{3-}$
- (c) $[\text{Co}(\text{CN})_6]^{3-}$
- (d) $[\text{NiCl}_4]^{2-}$

Case Study 8

Read the passage given below and answer the following questions :

Valence bond theory considers the bonding between the metal ion and the ligands as purely covalent. On the other hand, crystal field theory considers the metal-ligand bond to be ionic arising from electrostatic interaction between the metal ion and the ligands. In coordination compounds, the interaction between the ligand and the metal ion causes the five d -orbitals to split-up. This is called crystal field splitting and the energy difference between the two sets of energy level is called crystal field splitting energy. The crystal field splitting energy (Δ_o) depends upon the nature of the ligand. The actual configuration of complexes is divided by the relative values of Δ_o and P (pairing energy).

If $\Delta_o < P$, then complex will be high spin.

If $\Delta_o > P$, then complex will be low spin.

The following questions are multiple choice questions. Choose the most appropriate answer :

(i) Which of the following ligand has lowest Δ_o value?

- (a) CN^- (b) CO (c) F^- (d) NH_3

(ii) The crystal field splitting energy for octahedral (Δ_o) and tetrahedral (Δ_t) complex is related as

- (a) $\Delta_t = \frac{1}{2}\Delta_o$ (b) $\Delta_t = \frac{4}{9}\Delta_o$ (c) $\Delta_t = \frac{3}{5}\Delta_o$ (d) $\Delta_t = \frac{2}{5}\Delta_o$

(iii) On the basis of crystal field theory, the electronic configuration of d^4 in two situations : (i) $\Delta_o > P$ and

(ii) $\Delta_o < P$ are

- | (i) | (ii) |
|----------------------|------------------|
| (a) $t_{2g}^4 e_g^0$ | $t_{2g}^3 e_g^1$ |
| (b) $t_{2g}^3 e_g^1$ | $t_{2g}^4 e_g^0$ |
| (c) $t_{2g}^3 e_g^1$ | $t_{2g}^3 e_g^1$ |
| (d) $t_{2g}^4 e_g^0$ | $t_{2g}^4 e_g^0$ |

(iv) Using crystal field theory, calculate magnetic moment of central metal ion of $[\text{FeF}_6]^{4-}$.

- (a) 1.79 B.M. (b) 2.83 B.M. (c) 3.85 B.M. (d) 4.9 B.M.

OR

Electronic configuration of d -orbitals in $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ ion in an octahedral crystal field is

- (a) $t_{2g}^1 e_g^0$ (b) $t_{2g}^2 e_g^0$ (c) $t_{2g}^0 e_g^1$ (d) $t_{2g}^1 e_g^1$

Case Study 9

Read the passage given below and answer the following questions :

Metal carbonyl is an example of coordination compounds in which carbon monoxide (CO) acts as ligand. These are also called homoleptic carbonyls. These compounds contain both σ and π character. Some carbonyls have metal-metal bonds. The reactivity of metal carbonyls is due to (i) the metal centre and (ii) the CO ligands. CO is capable of accepting an appreciable amount of electron density from the metal atom into their empty π or π^* orbitals. These types of ligands are called π -acceptor or π -acid ligands. These interactions increases the Δ_o value.

The following questions are multiple choice questions. Choose the most appropriate answer :

(i) What is the oxidation state of metal in $[\text{Mn}_2(\text{CO})_{10}]$?

- (a) +1 (b) -1 (c) +2 (d) 0

(ii) Among the following metal carbonyls, the C – O bond order is lowest in

- (a) $[\text{Mn}(\text{CO})_6]^+$ (b) $[\text{Fe}(\text{CO})_5]$ (c) $[\text{Cr}(\text{CO})_6]$ (d) $[\text{V}(\text{CO})_6]^-$

OR

Which of the following can be reduced easily?

- (a) $\text{V}(\text{CO})_6$ (b) $\text{Mo}(\text{CO})_6$ (c) $[\text{Co}(\text{CO})_4]^-$ (d) $\text{Fe}(\text{CO})_5$

(iii) The oxidation state of cobalt in $\text{K}[\text{Co}(\text{CO})_4]$ is

- (a) +1 (b) +3 (c) -1 (d) 0

(iv) Structure of decacarbonyl manganese is

- (a) trigonal bipyramidal (b) octahedral (c) tetrahedral (d) square pyramidal.

Case Study 10

Read the passage given below and answer the following questions :

Werner, a Swiss chemist in 1892 prepared and characterised a large number of coordination compounds and studied their physical and chemical behaviour. He proposed that, in coordination compounds, metals possess two types of valencies, *viz.* primary valencies, which are normally ionisable and secondary valencies which are non-ionisable. In a series of compounds of cobalt (III) chloride with ammonia, it was found that some of the chloride ions could be precipitated as AgCl on adding excess of AgNO₃ solution in cold, but some remained in solution. The number of ions furnished by a complex in a solution can be determined by precipitation reactions. The measurement of molar conductance of solutions of coordination compounds helps to estimate the number of ions furnished by the compound in solution.

In these questions (Q. No. i-iv), a statement of assertion followed by a statement of reason is given. Choose the correct answer out of the following choices.

- (a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
- (b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.
- (c) Assertion is correct statement but reason is wrong statement.
- (d) Assertion is wrong statement but reason is correct statement.

The following questions are multiple choice questions. Choose the most appropriate answer :

- (i) **Assertion :** The complex $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ does not give precipitate with silver nitrate solution.
Reason : The given complex is non-ionisable.

(ii) **Assertion :** The complex $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ gives precipitate corresponding to 2 mol of AgCl with AgNO_3 solution.

Reason : It ionises as $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+ + \text{Cl}^-$.

OR

Assertion : $\text{CoCl}_3 \cdot 4\text{NH}_3$ gives 1 mol of AgCl on reacting with AgNO_3 , its secondary valency is 6.

Reason : Secondary valency corresponds to coordination number.

(iii) **Assertion :** 1 mol of $[\text{CrCl}_2(\text{H}_2\text{O})_4]\text{Cl} \cdot 2\text{H}_2\text{O}$ will give 1 mol of AgCl on treating with AgNO_3 .

Reason : Cl^- ions satisfying secondary valence will not be precipitated.

(iv) **Assertion :** $\text{CoCl}_3 \cdot 3\text{NH}_3$ is not conducting while $\text{CoCl}_3 \cdot 5\text{NH}_3$ is conducting.

Reason : The complex of $\text{CoCl}_3 \cdot 3\text{NH}_3$ is $[\text{CoCl}_3(\text{NH}_3)_3]$ while that of $\text{CoCl}_3 \cdot 5\text{NH}_3$ is $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$.

Case Study 11

Read the passage given below and answer the following questions :

Arrangement of ligands in order of their ability to cause splitting (Δ) is called spectrochemical series. Ligands which cause large splitting (large Δ) are called strong field ligands while those which cause small splitting (small Δ) are called weak field ligands. When strong field ligands approach metal atom/ion, the value of Δ_o is large, so that electrons are forced to get paired up in lower energy t_{2g} orbitals. Hence, a low-spin complex is resulted from strong field ligand. When weak field ligands approach metal atom/ion, the value of Δ_o is small, so that electrons enter high energy e_g orbitals rather than pairing in low energy t_{2g} orbitals. Hence, a high-spin complex is resulted from weak field ligands. Strong field ligands have tendency to form inner orbital complexes by forcing the electrons to pair up. Whereas weak field ligands have tendency to form outer orbital complex because inner electrons generally do not pair up.

In these questions (Q. No. i-iv), a statement of assertion followed by a statement of reason is given. Choose the correct answer out of the following choices.

- (a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
 - (b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.
 - (c) Assertion is correct statement but reason is wrong statement.
 - (d) Assertion is wrong statement but reason is correct statement.
- (i) **Assertion :** In tetrahedral coordination entity formation, the d orbital splitting is inverted and is smaller as compared to the octahedral field splitting.
Reason : Spectrochemical series is based on the absorption of light by complexes with different ligands.
- (ii) **Assertion :** In high spin situation, configuration of d^5 ions will be $t_{2g}^3 e_g^2$.
Reason : In high spin situation, pairing energy is less than crystal field energy.

OR

- Assertion :** F^- ion is a weak field ligand and forms outer orbital complex.
Reason : F^- ion cannot force the electrons of d_{z^2} and $d_{x^2-y^2}$ orbitals of the inner shell to occupy d_{xy} , d_{yz} and d_{zx} orbitals of the same shell.
- (iii) **Assertion :** The crystal field model is successful in explaining the formation, structures, colour and magnetic properties of coordination compounds.
Reason : In spectrochemical series, ligands are arranged in a series of increasing field strength.
- (iv) **Assertion :** NF_3 is a weaker ligand than $N(CH_3)_3$.
Reason : NF_3 ionizes to give F^- ions in aqueous solution.

Case Study 12

Read the passage given below and answer the following questions :

Ligands are atoms or ions which can donate electrons to the central atoms. Ligands can be monodentate, bidentate or polydentate as well. Few ligands can coordinate with the central atom through more than one site, these are called ambidentate ligands. When a di- or polydentate ligand uses its two or more donor atoms to bind a single metal ion, it is said to be a chelating ligand.

In these questions (Q. No. i-iv), a statement of assertion followed by a statement of reason is given. Choose the correct answer out of the following choices.

- (a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
- (b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.
- (c) Assertion is correct statement but reason is wrong statement.
- (d) Assertion is wrong statement but reason is correct statement.

(i) **Assertion :** Glycinate ion is an example of monodentate ligand.

Reason : Glycinate contains N and O as donor atoms.

OR

Assertion : EDTA forms complex with divalent metals of 3d-series in the ratio of 1 : 1.

Reason : EDTA has 4 —COOH groups.

(ii) **Assertion :** Oxalate ion is a bidentate ligand.

Reason : Oxalate ion has two donor atoms.

(iii) **Assertion :** A chelating ligand must possess two or more lone pairs at such a distance that it may form suitable strain free 5 and 6 membered rings with the metal ion.

Reason : $\text{H}_2\text{N}-\text{NH}_2$ is a chelating ligand.

(iv) **Assertion :** In Zeise's salt coordination number of Pt is five.

Reason : Ethene is a monodentate ligand.

Case Study 13

Read the passage given below and answer the following questions :

For understanding the structure and bonding in transition metal complexes, the magnetic properties are very helpful. Low spin complexes are generally diamagnetic because of pairing of electrons, whereas high spin complexes are usually paramagnetic because of presence of unpaired electrons. Larger the number of unpaired electrons, stronger will be the paramagnetism. However magnetic behaviour of a complex can be confirmed from magnetic moment measurement. Magnetic moment $\mu = \sqrt{n(n+2)}$ B.M. where n = number of unpaired electrons. Greater the number of unpaired electrons, more will be the magnetic moment.

In these questions (Q. No. i-iv), a statement of assertion followed by a statement of reason is given. Choose the correct answer out of the following choices.

- (a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
- (b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.
- (c) Assertion is correct statement but reason is wrong statement.
- (d) Assertion is wrong statement but reason is correct statement.

(i) **Assertion :** Both $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ have same magnetic moment.

Reason : Number of unpaired electrons in Cr^{2+} and Fe^{2+} are same.

(ii) **Assertion :** $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ is paramagnetic.

Reason : The Fe in $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ has three unpaired electrons.

(iii) **Assertion :** $[\text{Co}(\text{en})_3]^{3+}$ is paramagnetic.

Reason : It is an inner orbital complex.

(iv) **Assertion :** $[\text{Ni}(\text{CO})_4]$ is diamagnetic and tetrahedral in shape.

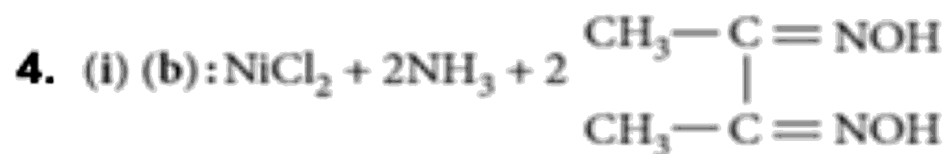
Reason : $[\text{Ni}(\text{CO})_4]$ contains no unpaired electrons and involves dsp^2 hybridisation.

OR

Assertion : $[\text{Ni}(\text{CN})_4]^{2-}$ is diamagnetic complex.

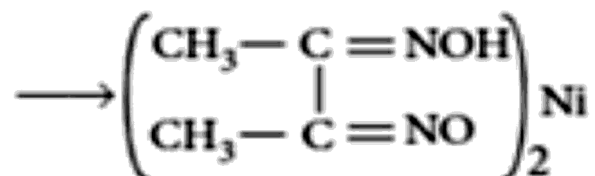
Reason : It involves dsp^2 hybridisation and there is no unpaired electron.

HINTS & EXPLANATIONS



OR

(b) : Ligand NO_2^- is ambidentate ligand as it can donate electrons through either nitrogen (NO_2) or oxygen



(ii) (c)

(iii) (b)

OR

(d) : It has no unpaired electrons hence, it is diamagnetic.

(iv) (d): (a) and (b) are coordination compounds hence cannot give free Fe^{2+} or Fe^{3+} ions in solution.

(c) and (d) represent simple compounds hence are free to give ions in solution, but only $\text{Fe}_2(\text{SO}_4)_3$ contains Fe^{3+} ions. $(\text{NH}_4)_2\text{SO}_4 \cdot \text{FeSO}_4 \cdot 6\text{H}_2\text{O}$ contains Fe^{2+} ions not Fe^{3+} ions.

5. (i) (c): Ligands are named in alphabetical order irrespective of their charge.

(ii) (b)

(iii) (d)

(iv) (d)

(ONO).

6. (i) (c): (A) : sp^3d^2 hybridisation (outer orbital)

No. of unpaired electrons = 4

(B) : d^2sp^3 hybridisation (inner orbital)

No. of unpaired electrons = 0

(C) : sp^3d^2 hybridisation (outer orbital)

No. of unpaired electrons = 5

(D) : d^2sp^3 hybridisation (inner orbital)

No. of unpaired electron = 1

(ii) (c) : It has 5 unpaired electrons.

OR

(c) : As H_2O is a weak ligand so, it should be $t_{2g}^3 e_g^2$

(iii) (a): Magnetic moments of (A), (B), (C) and (D) are respectively

$$\sqrt{4(4+2)}, 0, \sqrt{5(5+2)}, \sqrt{1(1+2)}$$

(iv) (c)

7. (i) (d): It involves sp^3d^2 hybridisation and not d^2sp^3 .

(ii) (b) : Magnetic moment of 3.83 B.M. suggests that it has 3 unpaired electrons,

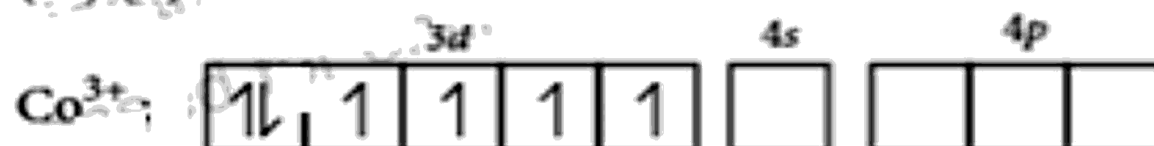
$$\therefore n = 3 \text{ i.e., } \text{Cr}^{3+} : 3d^3$$

It involves d^2sp^3 hybridisation so correct distribution of electrons is $3d_{xy}^1, 3d_{yz}^1, 3d_{zx}^1$.

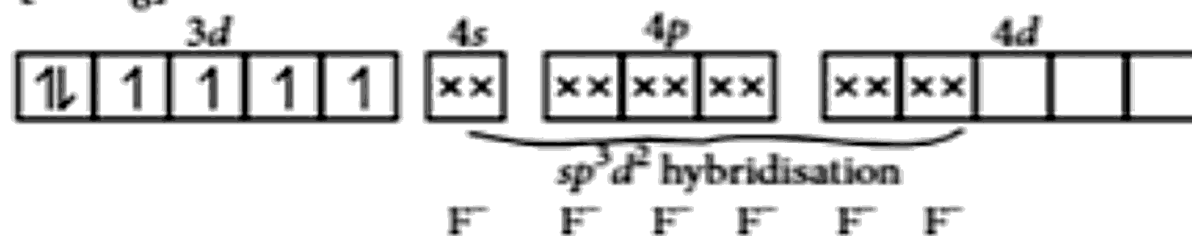
OR

(d) : $[\text{Co}(\text{NH}_3)_6]^{3+}$ is d^2sp^3 hybridised with all electrons paired hence, it is diamagnetic and inner orbital complex.

(iii) (b):

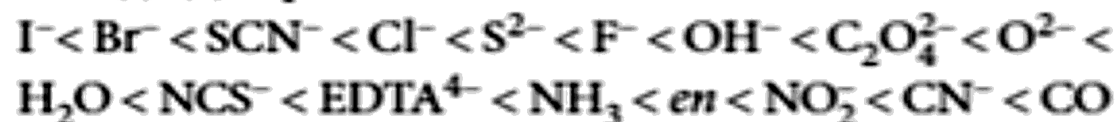


$[\text{CoF}_6]^{3-}$:



(iv) (c) : Inner orbital complexes are formed with strong ligands as they force electrons to pair up and hence the complex will be either diamagnetic or will have less number of unpaired electrons.

8. (i) (c) : Spectrochemical series :



(ii) (b)

(iii) (a) : When $\Delta_o > P$, the electrons paired up in the t_{2g} level rather than going to the e_g level, so

when $\Delta_o > P : t_{2g}^4 e_g^0$

and $\Delta_o < P : t_{2g}^3 e_g^1$

(iv) (d) : $\text{Fe}^{2+} : 3d^6 \Rightarrow t_{2g}^4 e_g^2$

(Since, F^- is a weak field ligand)

Hence four unpaired electrons are present.

Magnetic moment (μ)

$$= \sqrt{n(n+2)} = \sqrt{4(4+2)} = 4.9 \text{ B.M.}$$

OR

(a) : In $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$, Ti is in +3 oxidation state and there is only one electron in d -orbital.

9. (i) (d) : Oxidation state of Mn in $[\text{Mn}_2(\text{CO})_{10}]$ is zero.

(ii) (d) : In $[\text{V}(\text{CO})_6]^-$, the anionic carbonyl complex can delocalise more electron density to antibonding

π -orbital ($d\pi$ - $p\pi$ back bonding) of CO and thus lowers the bond order.

OR

(a) : $V(CO)_6$ can be easily reduced to $[V(CO)_6]^-$. $V(CO)_6$ has a total of 17 bonding electrons, hence it is very reactive and unstable. $[V(CO)_6]^-$ on the other hand has complete set of 18 bonding electrons as an electron is added into the bonding orbital when $V(CO)_6$ gets reduced to $[V(CO)_6]^-$. All others have 18 bonding electrons.

(iii) (c) : $K[Co(CO)_4]$
 $+1 + (x) + 4(0) = 0$ or $x = -1$

(iv) (d): $Mn_2(CO)_{10}$ is made up of two square pyramidal $Mn(CO)_5$ units joined by Mn – Mn bond.

10.(i) (a)

(ii) (d): $[Co(NH_3)_4Cl_2]Cl + AgNO_3 \longrightarrow$
 $[Co(NH_3)_4Cl_2]^+ + AgCl\downarrow$

Thus it gives precipitate of 1 mol of AgCl.

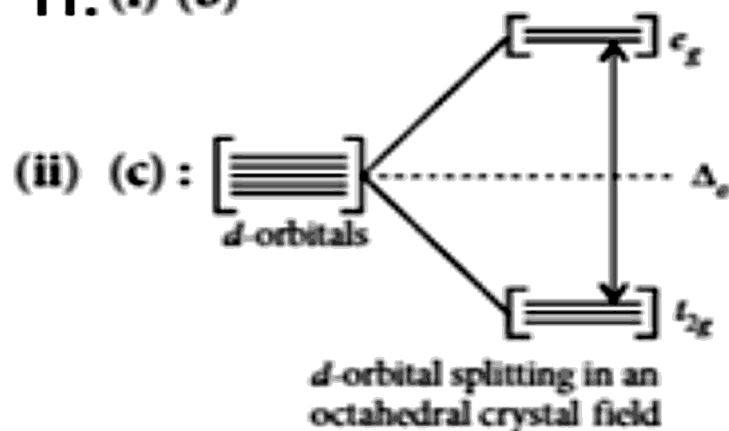
OR

(b) : $CoCl_3 \cdot 4NH_3$ gives 1 mol AgCl on reaction with $AgNO_3$, hence the complex can be represented as $[CoCl_2(NH_3)_4]Cl$.

(iii) (a): The Cl^- ions outside the coordination sphere can only be precipitated.

(iv) (a)

11. (i) (b)



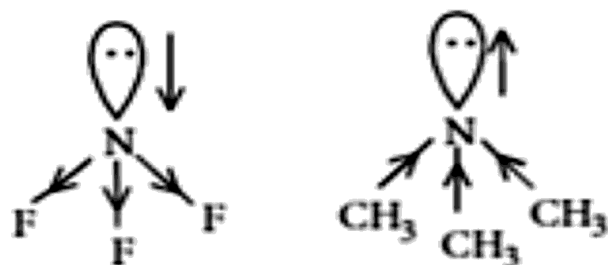
$\therefore$ In high spin situation, $\Delta_o < P$, in d^5 configuration, 4th and 5th electron are added to e_g rather than t_{2g} . So, configuration of d^5 ion will be $t_{2g}^3 e_g^2$.

OR

(a)

(iii) (b)

(iv) (c): Due to high electronegativity of F-atoms, the lone pair of N-atom in NF_3 molecule cannot be ligated easily. Whereas in $\text{N}(\text{CH}_3)_3$, CH_3 group is an electron releasing group, thus lone pair of N-atom in $\text{N}(\text{CH}_3)_3$ molecule can be ligated easily.



Except, nitrogen fluoride, all other halides hydrolyse in water.

12. (i) (d): Glycinate ion is an example of bidentate ligand. It contains N and O as donor atoms.

OR

(b) : EDTA is a hexadentate ligand. It forms complex with central metal in the ratio 1 : 1 in which it binds through two nitrogen atoms and four oxygen atoms.

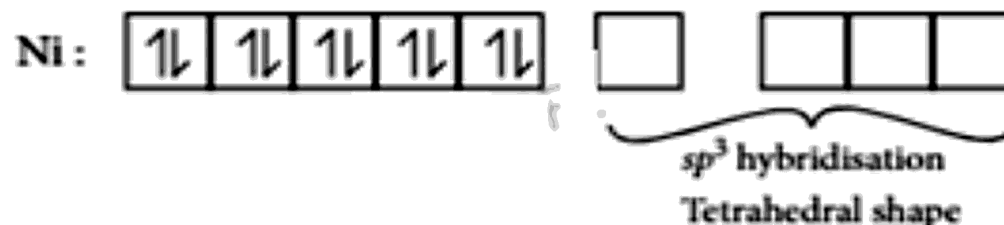
(ii) (a)

(iii) (c) : $\text{H}_2\text{N} - \text{NH}_2$ does not act as chelating ligand. The coordination by hydrazine leads to a three member highly unstable strained ring and thus it does not act as chelating agent.

(iv) (d): In Zeise's salt, coordination no. of Pt is 4. Ethylene is a monodentate ligand.

13. (i) (a): Spin only magnetic moment, $\mu = \sqrt{n(n+2)}$

CO is strong ligand thus,



OR

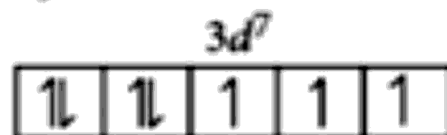
(a)

where n = number of unpaired electrons.

As the number of unpaired electrons in Cr^{2+} ($[\text{Ar}]3d^4$) and Fe^{2+} ($[\text{Ar}]3d^6$) are same, hence $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ will have same magnetic moment.

(ii) (a) : $\text{Fe}^+ : [\text{Ar}] 3d^6 4s^1$

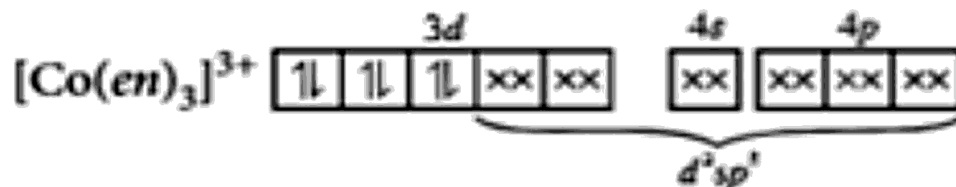
When the weak field ligand H_2O and strong field ligand NO^+ attack, the configuration changes as follows : $\text{Fe}^+ : [\text{Ar}] 3d^7 4s^0$



$\therefore \text{Fe}^+$ has 3 unpaired electrons.



In presence of strong ethylenediamine ligand the electrons get paired.



Thus inner orbital complex with no unpaired electrons.

(iv) (c) : $[\text{Ni}(\text{CO})_4]$ contains $\text{Ni}(0)$.

